Machine Learning 2

Super AI Engineer Season 6

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CLASSIFICATION ANALYSIS IN DATA SCIENCE

PROJECT OVERVIEW

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- In this hands-on project, we will train several classification algorithms namely Logistic Regression, Support Vector Machine, K-Nearest Neighbors, and Random Forest Classifier to predict the churn rate of Telecommunication Customers.
- Telecom service providers use customer attrition analysis as one of their key business metrics because the cost of retaining an existing customer is far less than acquiring a new one.
- Data Science empower companies by analyzing customer attrition rate based on several factors which includes various services subscribed by the customers, tenure rate, gender, senior citizen, payment method, etc.

INPUTS AND OUTPUTS

INPUTS

PAYMENT METHOD
TENURE
SUBSCRIBED SERVICES
TOTAL MINUTES
STATE
AREA CODE
INTERNATIONAL PLAN
DAY MINUTES
NIGHT MINUTES

CLASSIFIER MODEL

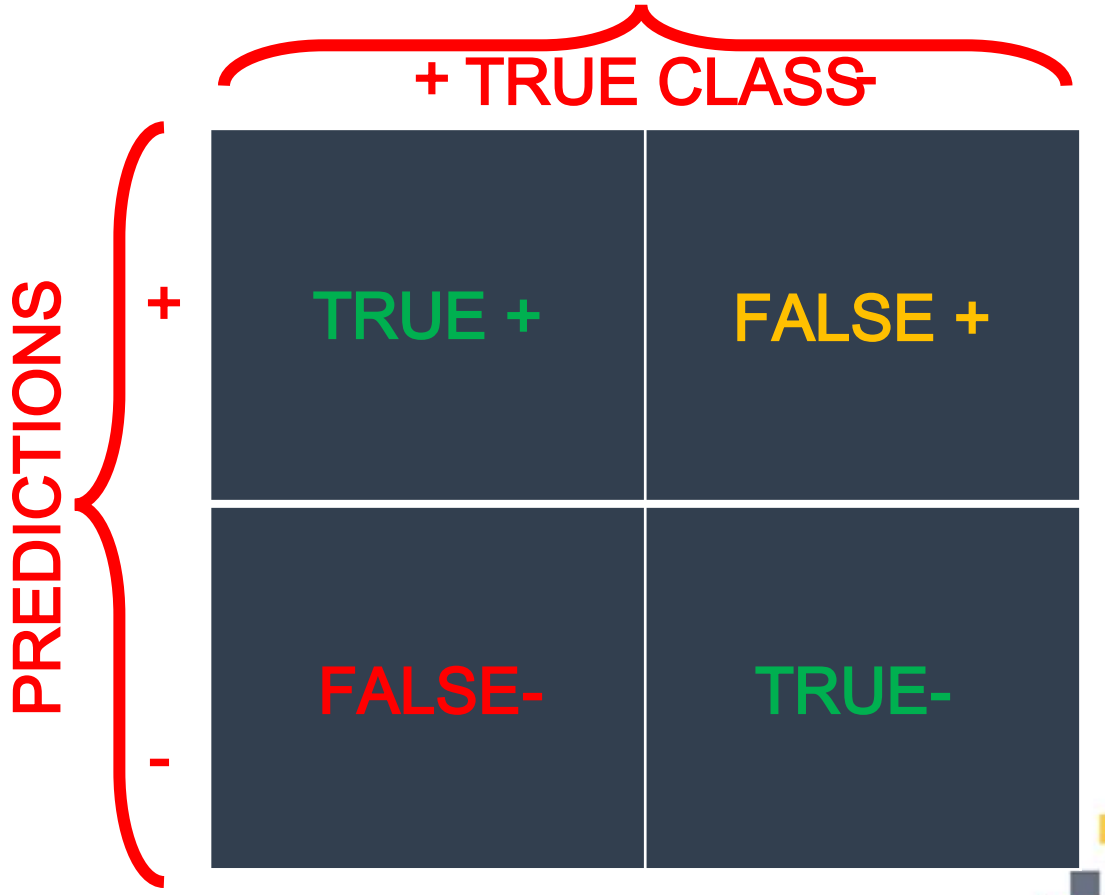
OUTPUT

ATTRITION
(0 or 1)

CLASSIFICATION MODELS KPIs

CLASSIFICATION MODEL KPIs

- Classification Accuracy = (TP+TN) / (TP + TN + FP + FN)
- Misclassification rate (Error Rate) = (FP + FN) / (TP + TN + FP + FN)
- **Precision** = TP/Total TRUE Predictions = TP/ (TP+FP) (When model predicted TRUE class, how often was it right?)
- **Recall** = TP/ Actual TRUE = TP/ (TP+FN) (when the class was actually TRUE, how often did the classifier get it right?)



DIABETES EXAMPLE

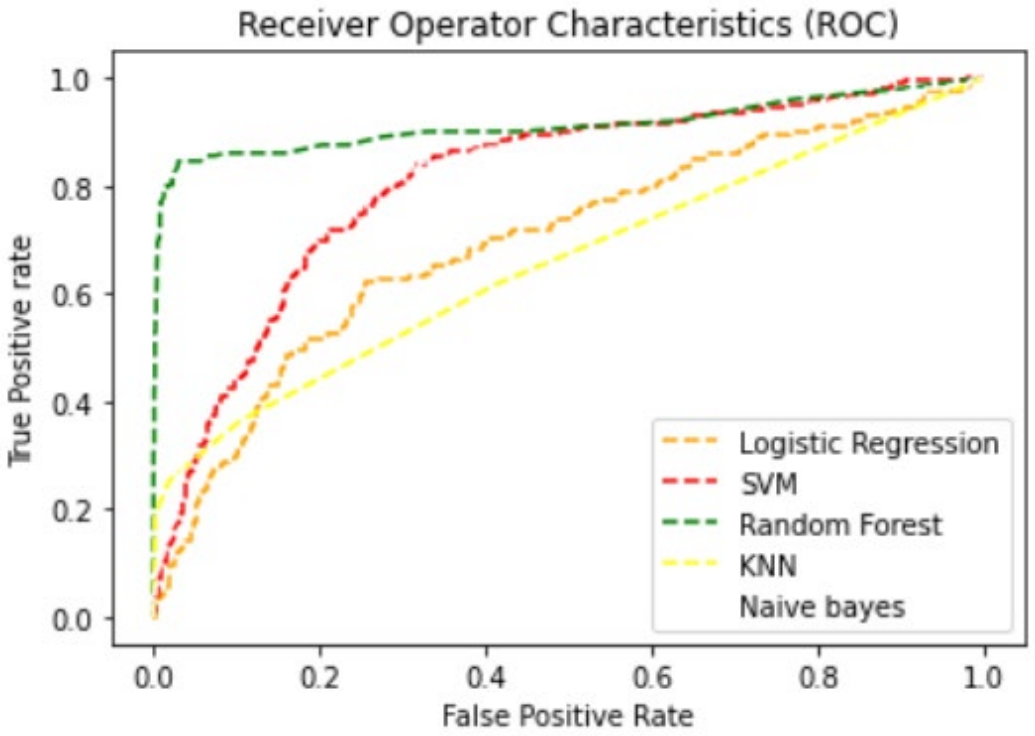
PREDICTIONS \ TRUE CLASS	+	-
+	TP = 1	FP = 1
-	FN = 8	TN = 90

FACTS:
100 PATIENTS TOTAL
91 PATIENTS ARE HEALTHY
9 PATIENTS HAVE DIABETES

- Accuracy is generally misleading and is not enough to assess the performance of a classifier.
- Recall is an important KPI in situations when dataset is highly unbalanced; cases when you have small diabetes patients compared to healthy ones.

- Classification Accuracy = (TP+TN) / (TP + TN + FP + FN) = 91%
- Precision = TP/Total TRUE Predictions = TP/ (TP+FP) = ½ =50%
- Recall = TP/ Actual TRUE = TP/ (TP+FN) = 1/9 = 11%

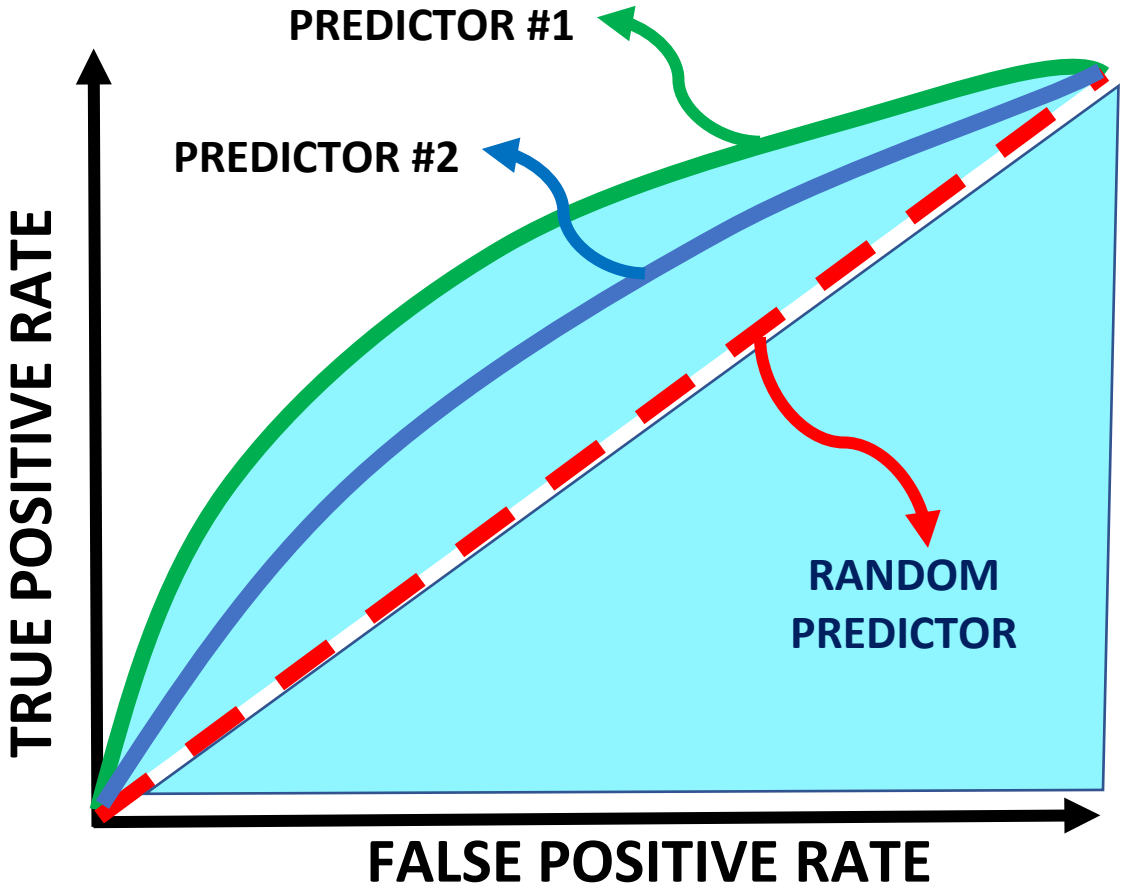
ROC (RECEIVER OPERATING



- ROC Curve is a metric that assesses the model ability to distinguish between binary (0 or 1) classes.
- ROC curve is created by plotting the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings.
- Points above the diagonal line represent good classification (better than random)
- The model performance improves if it becomes skewed towards the upper left corner.

Photo Credit:
https://commons.wikimedia.org/wiki/File:Roccurves.png

AUC (AREA UNDER CURVE)



- The light blue area represents the area Under the Curve of the Receiver Operating Characteristic (AUROC).
- The diagonal dashed red line represents the ROC curve of a random predictor with AUROC of 0.5.
- If ROC AUC = 1, this means perfect classifier
- The higher the AUC, the better the model is at predicting 0s as 0s and 1s as 1s.

Predictor #1 is better than predictor #2

CODE DEMO

CODE DEMO

Classification in DS.ipynb

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RAM Disk Editing

..
sample_data
ROC.png
telecom_churn.csv

CODING TASK #1: IMPORT LIBRARIES/DATASETS AND PERFORM EXPLORATORY DATA ANALYSIS

```
import numpy as np # Multi-dimensional array manipulation
import pandas as pd # DataFrame Manipulation
import matplotlib.pyplot as plt # Data Visualization
import seaborn as sns # Data Visualization
from sklearn import metrics
```

```
[2] # Read the CSV file
    telecom_df = pd.read_csv("telecom_churn.csv")
```

```
[3] # Load the top 5 instances
    telecom_df.head()
```

	state	account_length	area_code	phone_number	international_plan	voice_mail_plan	number_vmail_messages	total_day_minutes	total_day_calls	total_da
0	16	128	415	2845	0	1	25	265.1	110	
1	35	107	415	2301	0	1	26	161.6	123	
2	31	137	415	1616	0	0	0	243.4	114	
3	35	84	408	2510	1	0	0	299.4	71	
4	36	75	415	155	1	0	0	166.7	113	

FINAL CAPSTONE PROJECT

PROJECT OVERVIEW

- Assume that you work as a data scientist at a bank in Taiwan.
- The bank has collected extensive data about its customers such as demographics, historical payments record, amount of bill dollar values.
- Data has been collected between April 2005 to September 2005.
- The data consists of 25 variables. Let's explore these variables in the next slide!
- Data Source: https://www.kaggle.com/uciml/default-of-credit-card-clients-dataset

INPUTS/OUTPUTS

OUTPUT:

- default.payment.next.month: Default payment (1=yes, 0=no)

INPUTS:

- ID: ID of each client
- LIMIT_BAL: Amount of given credit in NT (New Taiwan) dollars
- SEX: Gender (1=male, 2=female)
- EDUCATION: (1=graduate school, 2=university, 3=high school, 4=others, 5=unknown, 6=unknown)
- MARRIAGE: Marital status (1=married, 2=single, 3=others)
- AGE: Age in years
- PAY_0: Repayment status in Sep, 2005 (-1=pay duly,
- 1=payment delay for one month, 2=payment delay for two months, ... 8=payment delay for eight months, 9=payment delay for nine months and above)
- PAY_2: Repayment status in August, 2005 (scale same as above)
- PAY_3: Repayment status in July, 2005 (scale same as above)
- PAY_4: Repayment status in June, 2005 (scale same as above)
- PAY_5: Repayment status in May, 2005 (scale same as above)
- PAY_6: Repayment status in April, 2005 (scale same as above)
- BILL_AMT1: Amount of bill statement in September, 2005 (NT dollar)
- BILL_AMT2: Amount of bill statement in August, 2005 (NT dollar)
- BILL_AMT3: Amount of bill statement in July, 2005 (NT dollar)
- BILL_AMT4: Amount of bill statement in June, 2005 (NT dollar)
- BILL_AMT5: Amount of bill statement in May, 2005 (NT dollar)
- BILL_AMT6: Amount of bill statement in April, 2005 (NT dollar)
- PAY_AMT1: Amount of previous payment in September, 2005 (NT dollar)
- PAY_AMT2: Amount of previous payment in August, 2005 (NT dollar)
- PAY_AMT3: Amount of previous payment in July, 2005 (NT dollar)
- PAY_AMT4: Amount of previous payment in June, 2005 (NT dollar)
- PAY_AMT5: Amount of previous payment in May, 2005 (NT dollar)
- PAY_AMT6: Amount of previous payment in April, 2005 (NT dollar)

FINAL CAPSTONE PROJECT TASKS

Complete the following tasks:

1. Import the "UCI_Credit_Card.csv" dataset
2. Perform Exploratory Data analysis and visualization (feel free to pick any reasonable data visualizations)
3. Prepare the data for model training and split it into training and testing
4. Train and evaluate an XG-Boost classifier model
5. Train and evaluate a Support Vector Machine classifier model
6. Train and evaluate a Naive Bayes classifier model
7. Train and evaluate a logistic regression model
8. Train and evaluate a Random Forest classifier model
9. Train and evaluate a K-Nearest Neighbors classifier model
10. Plot the ROC curve for all classifier and calculate the AUC
11. Which model performs best?